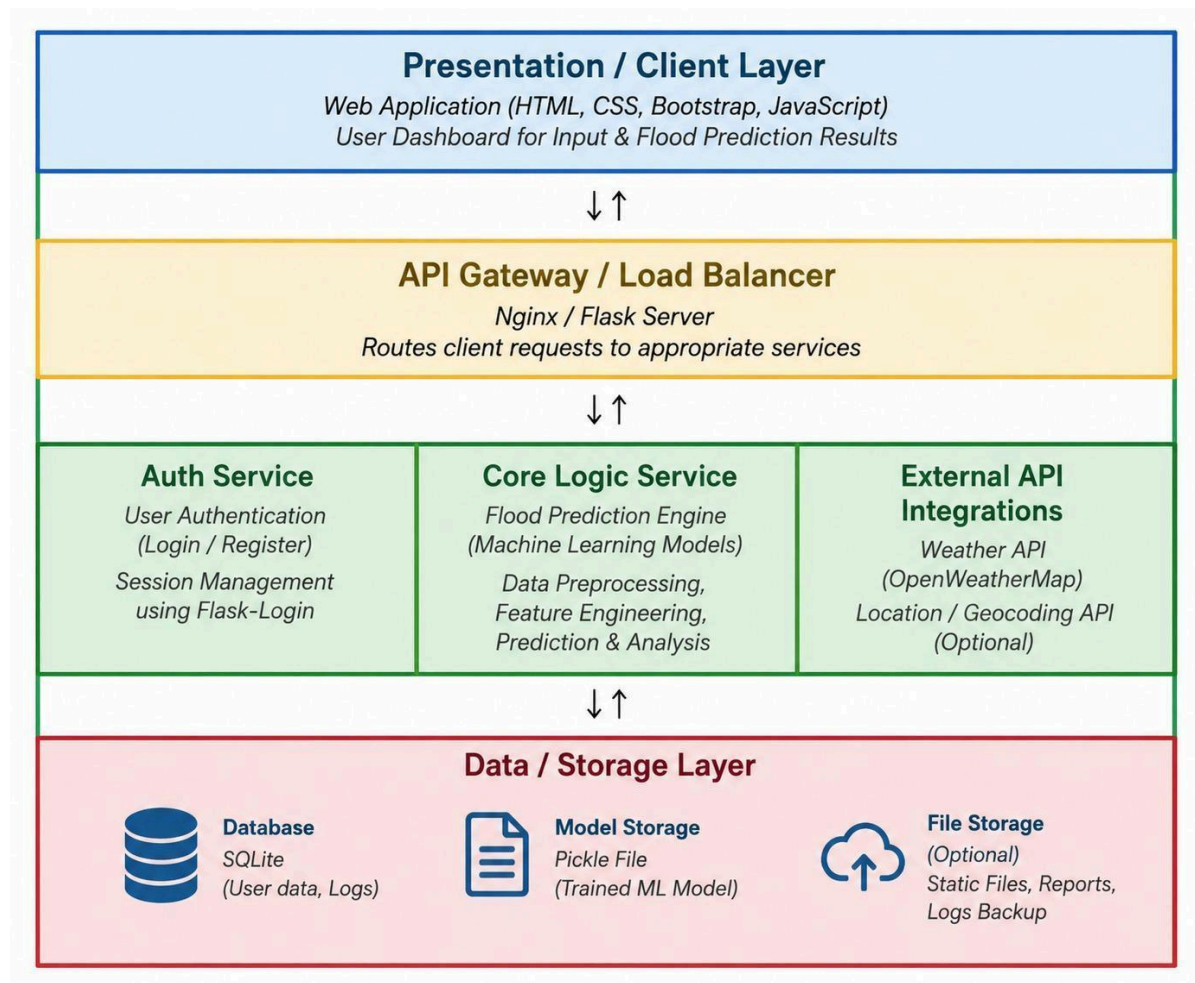


Solution Architecture

Date	28 June 2026
Team ID	
Project Name	Rising Waters-ML Flood Prediction System
Maximum Marks	5 Marks

Solution Architecture Diagram



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the user interface where users enter weather parameters and view flood prediction results.	HTML, CSS, JavaScript, Flask Templates
API Gateway/Application Layer	Handles HTTP requests, routes user inputs, validates data, and connects the frontend to the prediction service.	Flask (Python)
Authentication Module	Manages user login and session handling to allow secure access to the application.	Flask Session Management
Core Logic Service	Performs data preprocessing, feature scaling, loads the trained XGBoost model, and predicts flood probability.	Python, Scikit-learn, XGBoost, NumPy, Pandas, Joblib/Pickle
External Data Integration	Uses the Kaggle flood dataset for training and supports deployment on IBM Cloud.	Kaggle Dataset, IBM Cloud
Microservices	Implements the core flood prediction service, including data preprocessing, feature scaling, loading the trained XGBoost model, and generating flood predictions.	Implements the core flood prediction service, including data preprocessing, feature scaling, loading the trained XGBoost model, and generating flood predictions.
Database	Stores user information, weather data, trained model (<code>floods.save</code>), and prediction results.	SQLite/MySQL (depending on your implementation), Joblib/Pickle